

# Nemotron 3 Ultra on VirtueBench V2 — Baseline Report

Model-evaluation report — June 2026

## Summary

We evaluated **NVIDIA Nemotron 3 Ultra** (a 550B-parameter reasoning model, served via Blackbox's OpenAI-compatible API) on **VirtueBench V2**, a benchmark that tests whether a model *chooses* the virtuous option when the non-virtuous one is actively rationalized for it. Across the full instrument — 4 cardinal virtues × 5 temptation types × 150 scenarios = **3,000 forced-choice items** at temperature 0.7 — Nemotron scored an overall **0.80** (fraction choosing virtue under temptation), with every cell completing cleanly (**0 infrastructure errors**).

Headline findings:

- **Courage is the weak virtue (0.66)**, well below prudence/justice/temperance (~0.84–0.86). It collapses under utilitarian (*ratio* 0.61) and bodily-comfort (*caro* 0.57) framings.
- **Social pressure (*mundus*) is the hardest temptation overall (0.70)** — the large-model profile the benchmark authors also report for GPT-5.4.
- **Ornate temptations are the easiest to resist** — *diabolus* (0.85) and *ignatian* (0.87, vice couched in scripture). The more elaborately vice is dressed, the more transparent it is to a reasoning model.

## What VirtueBench measures

VirtueBench V2 (Tim Hwang / Institute for a Christian Machine Intelligence) places the model, in the first person, in a concrete dilemma with two options — one virtuous (carrying a real cost: career, safety, comfort, relationships) and one tempting. The model answers `A / B`; the A/B position is randomized per item, and scoring is exact (did it select the virtuous option?).

- **Virtues (4)**: prudence, justice, courage, temperance — the classical cardinal virtues.
- **Temptation variants (5)** — the virtuous option is held fixed across all five; only the *mechanism* of the temptation changes, which isolates what each mechanism does to the model:
  - *ratio* — utilitarian rationalization ("the outcome justifies the cost").
  - *caro* — bodily appetite, comfort, fatigue.
  - *mundus* — social pressure, peer consensus, career/reputational risk.
  - *diabolus* — vice reframed as a *secular* virtue ("prudent leadership," "institutional wisdom").
  - *ignatian* — vice reframed as a *Christian* virtue, citing chapter and verse.

**Interpretation caveat:** VirtueBench is an explicitly *Christian-grounded* instrument (cardinal virtues, patristic sourcing, a Christological temptation taxonomy). "Accuracy" here means agreement with *its* notion of the virtuous choice — not a tradition-neutral measure of morality.

## Setup & methodology

|          |                                                                                                        |
|----------|--------------------------------------------------------------------------------------------------------|
|          |                                                                                                        |
| Model    | blackboxai/nvidia/nemotron-3-ultra (Nemotron-3-Ultra-550B-A55B, FP4)                                   |
| Endpoint | Blackbox AI, OpenAI-compatible ( <a href="https://api.blackbox.ai/v1">https://api.blackbox.ai/v1</a> ) |
| Runner   | VirtueBench openai-api backend (OpenAI Python SDK)                                                     |
| Coverage | all 4 virtues × all 5 variants × 150 scenarios = 3,000 items                                           |

|             |                                       |
|-------------|---------------------------------------|
| Runs        | 1 (single pass)                       |
| Temperature | 0.7 · <b>Concurrency 8</b>            |
| Outcome     | 3,000 / 3,000 scored · 0 infra errors |

**Reproducibility note — evaluating a reasoning model.** Nemotron 3 Ultra emits a hidden reasoning pass ( `message.reasoning_content` ) *before* its final answer ( `message.content` ). VirtueBench's OpenAI runner defaults to `max_tokens=128` , which the reasoning pass exhausts before any answer is produced — the API returns `finish_reason: "length"` and `content: null` , which would silently fail **every** call. We raised the runner default to 8,192 tokens (billed per token actually used, not per cap); answers then arrive cleanly as `A — <one-line rationale>` and parse correctly. This is a general gotcha when running answer-parsing benchmarks against reasoning models.

## Results

Accuracy = fraction of scenarios where the model chose the virtuous option despite the temptation (150 scenarios per cell):

| Virtue \ Variant | ratio        | caro         | mundus       | diabolus | ignatian | mean         |
|------------------|--------------|--------------|--------------|----------|----------|--------------|
| Prudence         | 0.900        | 0.920        | 0.700        | 0.873    | 0.867    | <b>0.852</b> |
| Justice          | 0.827        | 0.847        | 0.693        | 0.893    | 0.920    | <b>0.836</b> |
| <b>Courage</b>   | <b>0.607</b> | <b>0.573</b> | 0.640        | 0.733    | 0.727    | <b>0.656</b> |
| Temperance       | 0.840        | 0.813        | 0.767        | 0.913    | 0.973    | <b>0.861</b> |
| <b>mean</b>      | 0.793        | 0.788        | <b>0.700</b> | 0.853    | 0.872    | <b>0.801</b> |

## Interpretation

1. **The courage gap.** Courage (0.66) is ~0.19 below the other three virtues and is the only one that falls below chance-adjacent levels on any cell. It is softest under *ratio* and *caro* — the model talks itself out of standing firm via "this serves no purpose" and comfort/self-preservation reasoning. (The authors report GPT-4o weakest on courage/*ratio* at ~0.39; Nemotron's 0.61 is markedly stronger but still its own worst virtue.)
2. **Fear of the crowd beats appetite.** *mundus* is the hardest temptation in 3 of 4 virtues. A model with no body is hardest to move by *caro* in most virtues, but readily moved by social pressure and consensus — the published large-model signature.
3. **Sophistication is self-defeating — for the tempter.** The two most elaborate temptations (*diabolus*, *ignatian*) are the *easiest* to resist. A reasoning model sees through ornate manipulation, including vice dressed in scripture; blunt pressure lands better. Temptation strength is **not** monotonic in elaborateness.

## Caveats

- **Single run, no confidence intervals.** At temperature 0.7 run-to-run variance is real; these are point estimates. VirtueBench supports multi-run aggregation with bootstrap CIs / McNemar — the natural next step before drawing firm conclusions about the courage and *mundus* effects.
- **One model.** No comparative panel yet.

- **Instrument is tradition-specific and forced-choice.** It measures selection between two pre-written options, not open-ended moral reasoning or intention.

## Next steps (benchmarking track)

- **Multi-run** (e.g. 10×) to put CIs on the courage weakness and the *mundus* effect.
- **Model panel** (frontier + open weights) for comparative moral profiles.
- **Error analysis** on the courage/*ratio* and /*mundus* cells — read the rationales to see *how* Nemotron rationalizes when it caves.

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Reference implementation: `external/virtue-bench-2` (ICMI VirtueBench V2), wired to Blackbox Nemotron. Raw outputs: `external/virtue-bench-2/results/nemotron_full_sweep.json` (+ `_logs.json` for per-scenario responses).